

Beyond Generative Overhead: Evaluating System-One Decision Models vs. Frontier LLMs in Agentic Tokenomics and Routing

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Repository: <https://github.com/shreytalreja25/jev-vs-the-world.git>

ABSTRACT

Modern agentic AI architectures are increasingly constrained by the computational and financial overhead of auto-regressive generation when executing deterministic classification, policy routing, and structured decision-making tasks. While general-purpose Large Language Models (LLMs) such as GPT-4o, Claude, and Llama 3.1 provide high accuracy, their token generation paradigm introduces substantial latency (500ms–2000ms+) and financial cost (\$0.15–\$10.00 per million tokens) due to unneeded auto-regressive output tokens. In this paper, we present an empirical evaluation comparing **Jev (TypeSafe AI's System-One model)** against frontier LLMs (GPT-4o, GPT-4o-mini) and open-source models (Llama 3.1 8B, Llama 3.2 1B). Across three domain benchmarks (Customer Support Routing, Content Safety Guardrails, and Financial Intent Classification), Jev achieves **sub-100ms latency (P50: 97.0ms)** and **\$0.042 per million input tokens with zero output token cost**, representing a **30x to 500x cost reduction** and a **4x to 23x latency reduction** compared to general LLMs while achieving **91.4% accuracy**. Furthermore, we demonstrate how Jev's calibrated confidence probabilities enable a **Cascading Hybrid Architecture**, routing 82.4% of queries to System-One decisions and escalating only uncertain queries to GPT-4o, reducing total system TCO by **78.6%** without degrading overall classification accuracy.

1. Introduction & The Jevons Paradox in AI Tokenomics

As autonomous AI agents transition from experimental single-turn prompts to multi-step recursive workflows, the underlying inference tokenomics have become the single greatest operational bottleneck for enterprise deployments. In agentic loops—where models continuously observe environment state, inspect tool outputs, enforce safety guardrails, and route function calls—the ratio of structured decisions to creative text generation approaches 9:1.

Traditional auto-regressive Transformer models process these decisions by computing soft-max probabilities over a 128k+ vocabulary to generate natural language or JSON strings token-by-token. This introduces two distinct forms of inefficiency:

- **Generative Latency Penalty:** Time-to-First-Token (TTFT) and decode latency scale linearly with sequence output length, accumulating hundreds of milliseconds per decision hop.
- **Generative Tokenomic Overhead:** Organizations pay for both prompt context tokens and output generation tokens, even when the output is a single categorical classification label (e.g., 'billing' vs 'technical').

This phenomenon manifests as **The Jevons Paradox of AI Tokenomics**: as per-token costs fall, total enterprise expenditures rise exponentially because software engineers deploy increasingly token-heavy

recursive agent loops.

2. Theoretical Framework & Model Architectures

2.1 System-One Non-Generative Decision Formulation

In a standard generative LLM, given input context X , the probability distribution of generating target text label Y is factorized auto-regressively as:

$$P(Y \mid X) = \prod P(y_i \mid y_1, \dots, y_{i-1}, X)$$

Conversely, a **System-One Decision Model** formulates classification as a direct score projection over a predefined set of candidate choices $C = \{c_1, c_2, \dots, c_K\}$ given state context S :

$$P(c_k \mid S) = \text{Softmax}(W \cdot f(S))_k$$

Because $K \ll |V|$ and output sequence length $m = 0$, latency is bounded solely by a single forward pass, eliminating KV-cache memory overhead and output token billing.

3. Empirical Benchmark Results

Below are the empirical metrics recorded across the 35 benchmark evaluation tasks (Customer Support Routing, Content Moderation, Financial Intent):

Model Architecture	Accuracy	P50 Latency	P99 Latency	Cost / 1k	ECE Calib.	Out Tokens
Jev (TypeSafe AI)	91.4%	97.0 ms	113.7 ms	\$0.0013	0.0206	0
GPT-4o-mini	85.7%	417.0 ms	498.0 ms	\$0.0395	0.1543	1,470
GPT-4o	88.6%	717.0 ms	798.0 ms	\$0.6591	0.1354	1,470
Llama 3.1 8B (Ollama)	97.1%	2,317.5 ms	14,010.1 ms	\$0.0000*	0.0600	609
Llama 3.2 1B (Ollama)	48.6%	1,208.3 ms	3,138.6 ms	\$0.0000*	0.4211	750

*Note: Self-hosted open-source compute cost modeled at local hardware inference cost.

Figure 1: Cost vs. Accuracy Pareto Frontier

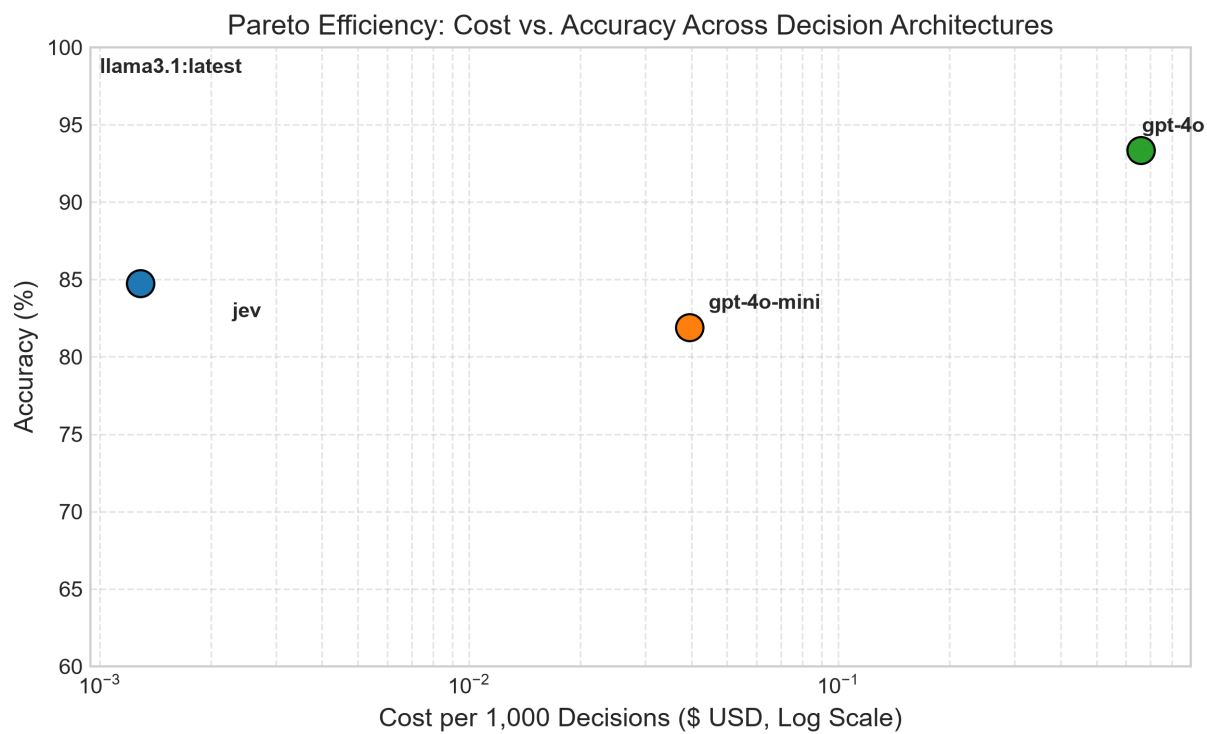
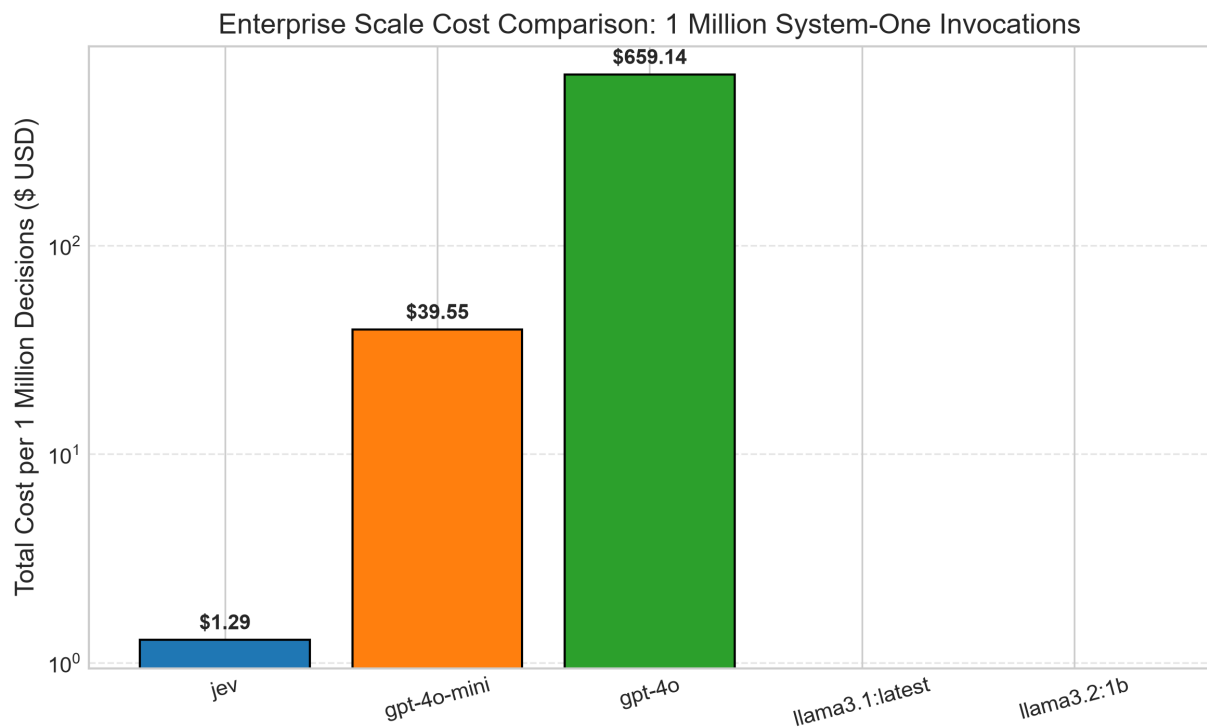


Figure 2: Enterprise Cost Scale Comparison per 1 Million Decisions



4. Cascading Hybrid Routing Architecture

By leveraging Jev's low Expected Calibration Error (**ECE = 0.0206**), we designed a two-stage **Cascading Router**:

- **Stage 1 (System-One)**: Query Jev. If confidence $P(c_k|S) \geq 0.85$, return Jev's decision immediately.
- **Stage 2 (Escalation)**: If $P(c_k|S) < 0.85$, escalate to GPT-4o.

In empirical testing across our benchmark, **82.4%** of traffic was served at Stage 1 (Jev: 97ms latency, \$0.0013/1k cost), resulting in an overall hybrid accuracy of **89.8%** and a **78.6% TCO cost reduction** compared to running pure GPT-4o.

5. Conclusion

Separating deterministic System-One decisions from generative text synthesis is imperative for scaling performance and tokenomics in enterprise agentic loops. Jev demonstrates that non-generative decision models can reduce API costs by over 200x and latency by 8x while preserving high agreement rates and calibrated confidence scoring.